



Sonu Kumar

Bachelor of Technology
Electrical Engineering
National Institute of Technology, Patna

+91-6299710356
sonuk.ug23.ee@nitp.ac.in
linkedin.com/in/SonuKumar
github.com/sumantpatel37819

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech., EE	National Institute of Technology, Patna	7.95	2023-2027
Senior Secondary	BSEB	91.2	2022
Secondary	CBSE	88.6	2020

INTELLECTUAL PROPERTY RIGHT (IPR)

- A Line-Marking Robot** [View](#)
An autonomous, high-precision line-marking robot
 - An autonomous, high-precision line-marking robot designed to automatically draw sports field and ground markings using coordinate-based navigation and sensor fusion.

EXPERIENCE

- Full-Stack Development Intern** 14 jan 2026 – 24 march 2026
VLED Lab, IIT Ropar / Ajasakha Reviewer System (Vinternship)
 - Developed Python-based web scraping and data processing pipelines to collect real-time agricultural market prices from state portals.
 - Integrated MCP servers and REST APIs to deliver structured market data to AI-powered agricultural assistants.

PROJECTS

- TourKaro – AI-Powered Tourist Guide Platform** Jul 2025 – Present
FastAPI, Node.js, MongoDB, Redis, Gemini API, ChromaDB, RAG, Docker, JWT, CI/CD [Website Link](#)
 - Engineered a production-grade AI travel platform using **RAG**, **vector embeddings**, **FAISS**, and **Gemini API** to deliver personalized destination recommendations and itinerary generation.
 - Developed scalable **FastAPI** and **Node.js** microservices with **JWT authentication**, REST APIs, and **Redis caching**, improving response latency and supporting concurrent user requests.
 - Containerized and deployed services using **Docker** with automated **CI/CD pipelines**, ensuring reliable and maintainable production releases.
- House Price Prediction Web App** [GitHub Repo](#)
Flask, Python, HTML/CSS/JS, Machine Learning
 - Built and deployed a Flask-based prediction service exposing **REST APIs** for real-time inference and model management.
 - Engineered a complete ML pipeline including data preprocessing, model training, evaluation, serialization, and API-based inference delivery.

SKILLS

- Programming:** Python, C++, C, JavaScript
- Backend & Databases:** FastAPI, REST APIs, MCP, LangChain RAG, MongoDB, FAISS, Redis
- Tools & Platforms:** Pandas, NumPy, PyTorch, Matplotlib, Git, GitHub, Docker, Linux, CI/CD
- Hardware & IoT:** ESP32, Arduino, Raspberry Pi, Jetson Nano

CERTIFICATIONS

- NIT Patna, Laser Bot Race Challenge** Jan 2024
- NPTEL Course, The Joy of Computing using Python** Jul-Oct 2025

ACHIEVEMENTS

- Ranked 3rd in district, Intermediate Science (2022)
- Won Gold in Laser Bot Race, Silver in Tinkering Lab, and Bronze in Grand Line Challenge

POSITIONS OF RESPONSIBILITY

- Vice President, Robotics Club, NIT Patna** 2025 – Present
Organized 5+ technical workshops and 2 annual flagship events, coordinating 200+ participants and managing a team of 10 volunteers.